

Advance Java & J2EE

Course Duration: 20Hrs

S. No	Content
1	MultiThreading • What is Thread • What is Multi-Tasking • How Multithreading works • Life Cycle of thread • Creating a thread: 1. Using Thread class 2. Using Runnable interface • What is Daemon Thread • How to create daemon thread • Synchronized Keyword • Converting a normal method as thread • Creating more than one thread • What is Garbage Collection
2	Collection Framework: • Introduction to collection • Use of Collections • Collection Object • Collection classes and interfaces • List Interface: 1. Introduction to List 2. How it stores the data in memory • Types:

	○ ArrayList ○ LinkedList ○ Stack ○ Vector ● Set Interface 1. Introduction to Set 2. How it stores the data in memory 3. Types: ○ HashSet ○ TreeSet ● Queue Interface 1. Introduction to Set 2. How it stores the data in memory 3. Types: ○ Deque ○ PriorityDeque
3	Map Interface 1. Introduction to Set 2. How it stores the data in memory 3. Types: ○ HashMap ○ TreeMap ○ LinkedHashMap ○ NavigatableMap
4	I/O Stream ● Introduction to I/O Stream ● What Input Stream ● What Output Stream ● Creating a file

	• Reading and Writing in a file • Closing and Deleting a file
5	Serialization & Deserialization • Introduction • Encrypting the data • Decrypting the data
6	Networking • What is networking • Network Topologies • What is Socket • Use of socket class • One way communication • Two way communication • Finding Inet Address
6	Java Swing • Swing introduction • Swing Classes • Creating a frame • GUI Application • Creating form, input, radio button, dropdown • Adding images in frame and button • What is Event Handling • How Events work
7.	Layout Manager • Introduction to Layout • Creating a frame with layout

	• Types of Layout: 1. Grid Layout 2. Box Layout 3. Card Layout 4. Flex Layout
8	JDBC • Java DataBase Connectivity Introsuction • MySQL Introduction • MySQL Administraction • Using SQL Queries. • Creating table and adding columns • Using Constrains in SQL • Inserting data, updating and deleting the tables. • Connecting Java to MySQL • CRUD – Create Read Update Delete Operation
9	Exercise & Projects.